

Open Data Portal Port Management



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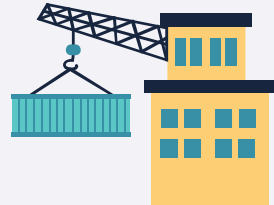
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Communication

Tools used for team communication



01 The Team



André Alves



Architect

Pedro Rei



DB master

Francisco Pinto



Project Manager

José Moreira



Advisor

Diogo Costa



Product Manager

Bruno Tavares



Devops

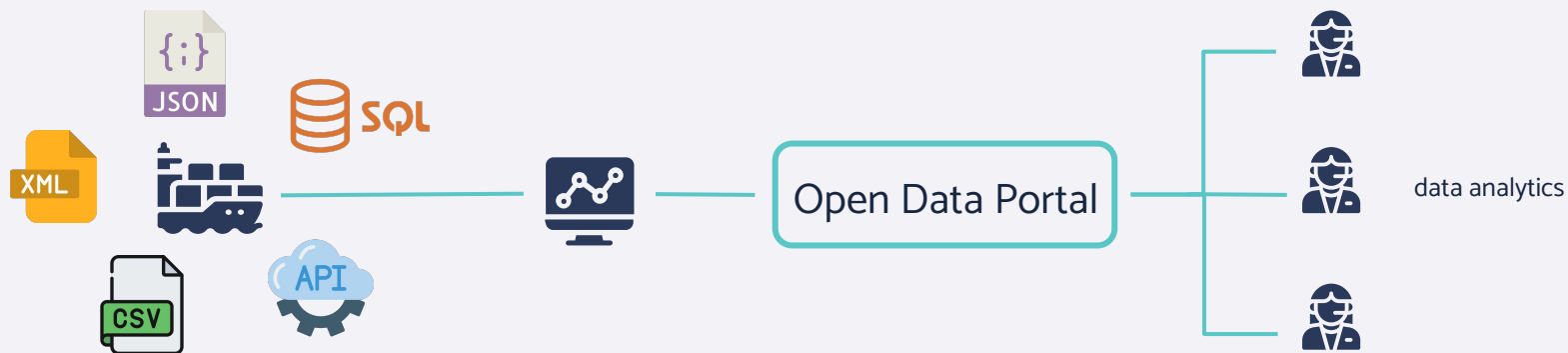
Paulo Pintor



Advisor

02 Context

The need for a single portal for port management data analytics



03

Problems



Multiple data source

Depending on who publishes the data the labels can differ, making difficult to gather statistics



Inconsistency in data points

E.g in cargo weight it can be in kg, tons or number of containers



Missing data

In some rows, there are important fields missing



Duplicated data

A large-scale database may contain duplicate data that leads to system overload and reduces operational efficiency



04

Goals



1

Use a machine learning algorithm to address data gaps



2

Perform outliers and inconsistencies detection in the data set

3

Develop a Data portal for data access

4

Data analytics for better comprehension of data set

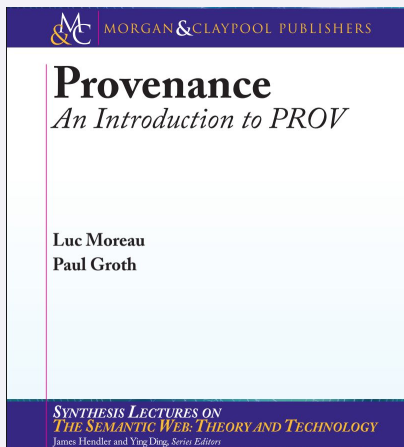


05

Related Work

James Cheney, Laura Chiticariu, and Wang-Chiew Tan. 2009. Provenance in Databases: Why, How, and Where. *Found. Trends databases* 1, 4 (April 2009), 379–474 - <https://doi.org/10.1561/19000000006>

Oracle. 2024. Data Deduplication. Retrieved February 20, 2025, from <https://www.oracle.com/uk/data-deduplication/>



Luc Moreau and Paul Groth. 2013. Provenance: An Introduction to PROV. Morgan & Claypool Publishers. Chapters 2 and 3.



05

Related Work on data provenance



Importance

In complex database systems like port operations management, data provenance ensures data integrity and transparency.



Why?

Identifies which input data contributed to query results, crucial for understanding data sources and their interrelationships.



How?

Describes the transformations data underwent to reach its current state, providing transparency and supporting process audits.



Where?

Traces the exact origin of data within the database, essential for validating data authenticity and accuracy



06

Expected Results



1

Create a well-structured database involves several key steps to ensure efficiency, scalability, and data integrity. With metadata attach to the documents.

2

Development of a website for easier data access, allowing data and metadata downloads.

3

Document analysis via graphs and tables, with the ability to track data provenance of documents, manual entry of data or created the use of a machine learning algorithm.



07

Timeline



Milestone 1

Requirement Gathering,
Goals, life cycle objectives,
calendar and Roles of Team



Milestone 3

Prototype, peer evaluation
and mid-term presentation
with supervisors



Milestone 2

Use Cases,
Architecture, Functional and
Nonfunctional Requirements



Milestone 4

Project presentation,
Posters, Video and Final
Deployment



07



Calendar

Week

Steps	1	2	3	4	5	6	7	8
Inception								
Definition of technologies to be used								
Uses cases								
Architecture design								
Explore the original database								
Creation of the database with metadata								
Implementation of the API								
Development of the website UI								



07



Calendar

Week

Steps	9	10	11	12	13	14	15	16	17
Development of the website UI									
Data prediction and outlier detection									
Prototype									
Metadata visualization									
Analysis module									
Finalization of the website UI									
Deployment									
Final Report									



08

Communication plan



Thanks!



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